

Yash Patel

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EDUCATION

- University of New Hampshire** Durham, New Hampshire
Bachelor of Science in Computer Engineering May 2026
Courses: FPGA Design, Embedded Systems Design, Digital Systems Design, Electronic Circuit Design, Communication Systems Design, Operating Systems, Computer Architecture

EXPERIENCE

- University of New Hampshire - Interoperability Labs** In Person - Part Time/Internship
Senior Test Engineer Apr. 2025 - Present
 - ISO 21111-6 Test Plan Optimization:** Contributed critical technical revisions and validation fixes to the OPEN Alliance Technical Committee, directly driving standardizations for upcoming ISO 21111-6 compliance frameworks.
 - Automotive Ethernet Processing and Reconstruction Tool:** Engineered a high-performance 100BASE-T1 signal processing engine in Rust to decode scrambled analog and digital PAM-3 signals, yielding a 30x processing speedup over legacy systems.
 - Single Pair Ethernet automation testing:** Applied Rust, MATLAB, and web technologies to develop a 100BASE-T1 automation testing tool capable of receiving captures and programmatically determining results without user intervention.
- Junior Test Engineer* Jan. 2024 – Apr. 2025
 - Development for custom FPGA test automation:** Architected an IEEE 802.3 Clause 22/45 MDIO transmitter in Verilog for a custom FPGA test automation framework, integrated into commercially available hardware validation boards.
 - Reservation Manager and Tester for 20+ bookings:** Managed the end-to-end testing lifecycle for concurrent vendor bookings, delivering detailed testing reports and analyzing Single-Pair Ethernet (SPE) transmissions.
 - Test Script Development:** Developed test scripts for an FPGA-based 100BASE-T1 test station to ensure accurate timing synchronization, enabling the Device Under Test (DUT) to properly execute suite requirements.

INDUSTRY CONTRIBUTIONS

- Contributor: OPEN Alliance IEEE 100BASE-T1 Test Suite V1.4:** Recognized in the specification acknowledgments for driving major technical revisions, including optimizing test coverage, deprecating obsolete tests, and refining functional requirements for conformance testing.

KEY PROFICIENCIES

- Languages:** Rust, C, Python, Bash/Shell, HTML/CSS/JS
- Hardware Description:** SystemVerilog, Verilog
- Protocols:** 100BASE-T1 (PHY-C, PCS), Ethernet, TCP/IP, CAN, I²C, SPI, UART, SSH
- Tools:** Wireshark, Oscilloscopes, Logic Analyzers, Packet Generators, Vivado, Git
- Platforms & OS:** GNU/Linux, Windows, STM32, ESP32, FPGA (Xilinx & Altera), AWS (IoT/S3)

PROJECTS

- Smart Adaptive Audio Equalizer Headset**
Senior Design Project
 - Collaborated with a team of 3:** to design an adaptive headset featuring active external (ambient) and internal (biological) noise cancellation.
 - Developed a Digital Signal Processing (DSP) calibration routine:** using a 60-second automated hearing test to create personalized audio profiles, dynamically mapping them over audio sources.
 - Engineered a dual-microphone feedback/feedforward system:** to eliminate city noise and internal interference (e.g., blood flow).
- MSP430 Morse Code Decoder:** Programmed an MSP430FR6989 to perform real-time text-to-Morse decoding, utilizing UART for input and GPIO-controlled LEDs and buzzers for signal output.
- Hyprhelp:** Engineered a Python-based utility for the Hyprland tiling window manager that parses system configuration files to generate dynamic, searchable keybinding menus.